

Near-field to Far-field Transformation of Cylindrical Scanning Antenna Measurement Using Two Dimension Fast-Fourier Transform

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Abstract— Radiation characteristics of an antenna should be tested through a precise measurement procedure. To provide an actual parameter of an antenna radiation, measurement must be performed in the far-field region and its distance is proportional to the antenna's dimension. Therefore, in a limited space of an anechoic chamber facility, antenna measurement has a constraint to a maximum dimension of the antenna under test. Near-field measurement method has been an alternative to overcome this problem. However, additional techniques, such as sampling, scanning mechanism, transformation computation, cannot be avoided to obtain the actual antenna radiation parameters. In this research, we study cylindrical scanning technique to obtain sampling data because it is simpler than the spherical scanning in mechanical setup and can produce radiation pattern results in completely two dimension polar diagrams. Our purpose is to provide low cost and reasonable performance of a near-field measurement system. Two dimension Fast-Fourier Transform is used to process and analyze the near-field data into the far-field data. Other methods, i.e. one dimensional Fast-Fourier transform and numerical method, are also performed to compare the accuracy and efficiency of the proposed method. Near-field data generated from experiment data measured of 8-array micro-strip antenna at resonant frequency 3.35 GHz. Transformation results show that average deviations of one dimension Fast-Fourier Transform, two dimension Fast-Fourier Transform and the numerical method are 6.83 dB, 3.04 dB, and 3.77 dB, respectively. These results show the minimum discrepancies between the transform and the expected far-field obtained by using the two dimensions Fast-Fourier Transform. This method has shown near-field measurement and transformation can predict the far-field radiation characteristics.

Keywords— *near-field; far-field; Fast-Fourier; cylindrical; transformation*

I. INTRODUCTION

Radiation pattern measurement can be divided into two techniques, i.e. direct and indirect [1]. The direct technique is referring to the antenna measurement conducted at a distance of far-field between antenna under test (AUT) and measurement points. This technique gives direct measurement results. However, some issues are coming for this measurement, such as attenuation fluctuation of weather condition and multipath, therefore, there are possibilities of inaccurate results [2].

Building an anechoic chamber can solve those issues, but it will not efficient if the antenna under test has large dimension because the larger size of the antenna under test, the more distance of measurement should be provided between both of them. Therefore, the second technique of indirect measurement has been studied to overcome those issues. This technique is referring to a measurement technique that is named as near field measurement technique. In this measurement, the observation points are located in the near field region or also known as the Fresnel zone. The measured data are then transformed into far-field by conducting some procedures of computation. This procedure can provide antenna radiation parameters, such as radiation pattern, beam width and gain [3].

More procedures and considerations are coming when the near field technique applied to measure an antenna than the direct far-field technique. The three main parts are scanning system, measurement equipment and computational mechanism [3]. There are three measurement area based on the general coordinate systems, i.e. planar, cylindrical and spherical [3]. Those three are increasing in complexity, accuracy and cost. In general, patterns in E-plane and H-plane are mostly considered in analysis, therefore, the cylindrical scanning is already able to provide those parameters and optimize the complexity and cost. Computation is the next step after conducting the near field measurement. In order to obtain accurate and efficient computation process, Fast Fourier Transform is able to be used in the computation [4].

Study of the near field measurement technique has been conducted over some decades [1]-[3], [5]-[13], [14]. Some researchers have studied in the planar scanning area [2, 6], the cylindrical scanning area [4], [12], [13], and also some vendors like SATIMO have provided spherical near field antenna system for research and commercial use [15]. In our research group, we have been studying and developing the near field method for planar [16]-[17] and cylindrical [18] scanning area. The two were chosen because its simplicity in mechanical and low cost to be developed. This paper will discuss measurement of an 8 element array antenna and a monopole antenna probe configuration by using a cylindrical scanning area. Three computations of FFT-1D, FFT-2D and numerical are conducted to find out the most accurate results in near- to far-field transformation.

II. NEAR FIELD MEASUREMENT

A. Radiation regions

Fig. 1 shows radiation regions from an antenna that consists of reactive near-field, radiating near-field or Fresnel region and far-field or Fraunhofer region. The difference of the three regions is in the field distribution. In the reactive near-field, the reactive fields are dominant that is the closest region of the antenna. The radiating near-field is a region between the reactive near-field and the far-field where angular field radiation is dependent to a distance from the antenna. The far-field region is a region where the angular radiation pattern independent to distance between the observation points and the antenna. Therefore, measurement of radiation pattern is generally conducted in the latter region. However, because a minimum distance should be fulfilled from formula $\frac{2D^2}{\lambda} + \lambda$ where D is antenna dimension and λ is wavelength, a problem is coming when the measurement conducted for a large antenna dimension and/or high resonant frequency. This condition leads to conduct the far-field measurement in a large space/area or to conduct measurement in a shorter distance with some additional analysis to obtain the desired far-field results.

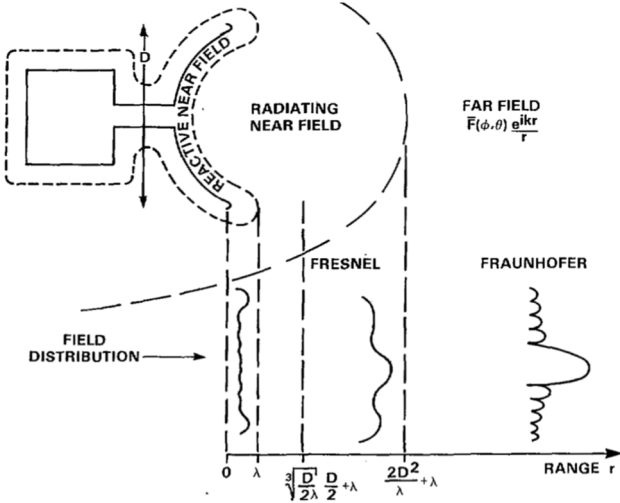


Fig. 1. Radiation regions of an antenna [6]

B. Near-field measurement

Near-field measurement means radiation measured in the radiating near-field region or also known as Fresnel region. The measured data consist of amplitude and phase of the radiation distribution after scanning by using a probe. The scanning area can form planar, cylindrical or spherical. Compared to the other two scanning areas, cylindrical scanning technique to obtain sampling data is simpler than the spherical scanning in mechanical setup and can produce radiation pattern results in completely two-dimensional polar diagram that not possible in the planar scanning. Fig. 2 shows a schematic of the cylindrical scanning system. It shows that the system can be developed by preparing an antenna under test rotated by a rotator and a probe antenna moved vertically from a one-line scanner. The distance in the vertical scanner and the angle of the rotatory movement

for two measurement points should be maintained not more than $\lambda/2$ and π/ka , respectively. The recommended distance between the antenna under test and the probe is $2 - 10 \lambda$ [9].

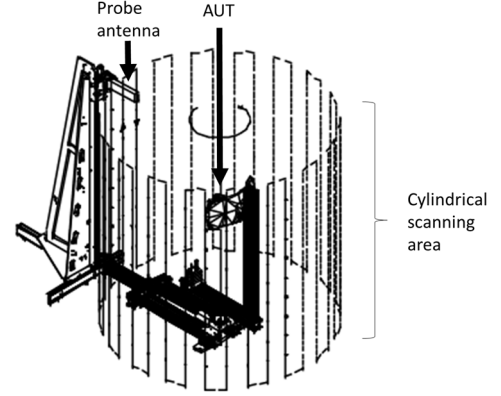


Fig. 2. A schematic of a cylindrical scanning system [7]

C. Near- to Far-field Transformation

E-field (E) and H-field (H) of electromagnetic waves radiated from an antenna in the far-field region can be represented by Equation (1) and (2) [3]. The two fields are as a function of elevation angles (θ) and azimuth angles (ϕ) of the spherical scanning system. Therefore, there are also two unit vectors in the fields, i.e. elevation unit vector ($\hat{\theta}$) and azimuth unit vector ($\hat{\phi}$). The E-field is able to be transformed to the H-field by dividing component of the E-field with impedance characteristic of the medium (η). Therefore, the two E-field components must be found to generate both E-field and H-field. The two components can be formulated in Equation (3) and (4) [12]. $a_n(h)$ and $b_n(h)$ are weighted amplitude of waves emitted by an antenna under test in cylindrical scheme represented in Equation (5) and (6). $c_m(-h)$, $c_m'(-h)$, $d_m(-h)$, and $d_m'(-h)$ are probe coefficient that used to calculate probe compensation [13]. In and In' in Equation 7 and 8 are functions that represent the output of the probe ($v(r, \phi, z)$ and $v'(r, \phi, z)$) when measure the antenna under test radiation. Some equations (Equation (3) – (8)) in a form that possible to be solved by using some calculation methods, such as DFT, FFT-1D, FFT-2D, and numerical computation.

$$E(\theta, \phi) = E_\theta(\theta, \phi)\hat{\theta} + E_\phi(\theta, \phi)\hat{\phi} \quad (1)$$

$$H(\theta, \phi) = \frac{1}{\eta} (E_\theta(\theta, \phi)\hat{\theta} - E_\phi(\theta, \phi)\hat{\phi}) \quad (2)$$

$$E_\theta(\theta, \phi) = j \sin \theta \sum_{n=-\infty}^{\infty} j^n b_n(h) \exp(jn\phi) \quad (3)$$

$$E_\phi(\theta, \phi) = \sin \theta \sum_{n=-\infty}^{\infty} j^n a_n(h) \exp(jn\phi) \quad (4)$$

$$a_n(h) = \frac{k^2}{\Lambda^2 \Delta_n(h)} (I_n(h) \sum_{m=-\infty}^{\infty} d_m'(-h) H_{n+m}^{(2)}(\Lambda r_0) - I_n'(h) \sum_{m=-\infty}^{\infty} d_m(-h) H_{n+m}^{(2)}(\Lambda r_0)) \quad (5)$$

$$b_n(h) = \frac{k^2}{\Lambda^2 \Delta_n(h)} (I_n'(h) \sum_{m=-\infty}^{\infty} c_m(-h) H_{n+m}^{(2)}(\Lambda r_0) - I_n(h) \sum_{m=-\infty}^{\infty} c_m'(-h) H_{n+m}^{(2)}(\Lambda r_0)) \quad (6)$$

$$I_n(h) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} v(r_0, \phi, z) \exp(-jn\phi) \exp(jhz) d\phi dz \quad (7)$$

$$I_n'(h) = \int_{-\infty}^{\infty} \int_{-\pi}^{\pi} v'(r_0, \phi, z) \exp(-jn\phi) \exp(jhz) d\phi dz \quad (8)$$

III. MEASUREMENT SETUP AND EXPERIMENT

Fig. 3 shows the measurement setup of cylindrical near-field measurement system that conducted in an anechoic chamber of Telecommunication Laboratory, Department of Electrical Engineering, Faculty of Engineering, Universitas Indonesia. In this experiment, a monopole antenna and an 8 array element microstrip array antenna at resonant frequency 3.35 GHz are used as probe antenna and antenna under test (AUT), respectively. The distance between the antenna under test and the probe antenna is 0.3 m where is in the Fresnel region. The distance and the angle between two points are 4 cm and 10° , respectively.

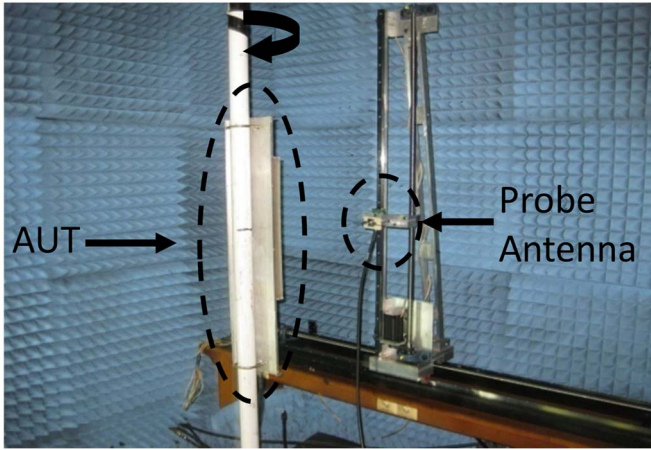


Fig. 3. Measurement setup of cylindrical near-field measurement

After the measurement setup was completed, the next is conducting measurement in the determined locations. The radiation is measured in two conditions of the probe; vertically and horizontally. After finishing the measurement, the data are transformed by Equations as mentioned in the previous part and using computation algorithms to solve the problems. The

transformed results are compared with the direct far-field measurement results to validate the experiments.

IV. RESULT AND DISCUSSION

Fig. 4 shows radiation pattern measurement results obtained from far-field and near-field measurements. It shows that the main lobe pattern from the transformation by using FFT-2D and numerical are closer to the far-field measurement result than transformation by using FFT-1D. In the side lobe direction around -40° , the FFT-2D has the lowest discrepancies than the others.

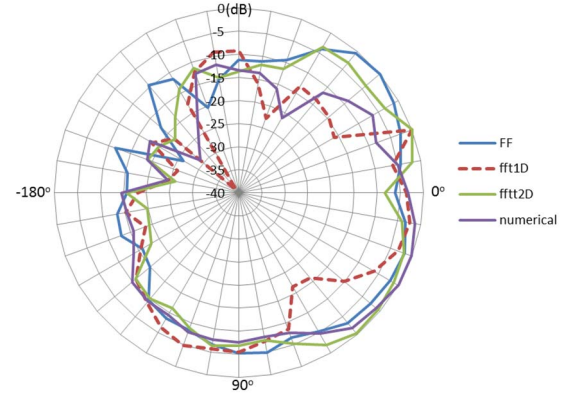


Fig. 4. Radiation patterns obtained from far-field and near field measurement at plane $\theta = 90^\circ$

Fig. 5 shows different plots between far-field and near-field measurement results. It shows the smallest differences in a range of angles of observation occur at the FFT-2D. By taking an average number of each calculation method, transformation results show that average deviations of one dimension Fast-Fourier Transform, two dimensions Fast-Fourier Transform and the numerical method are 6.83 dB, 3.04 dB, and 3.77 dB, respectively.

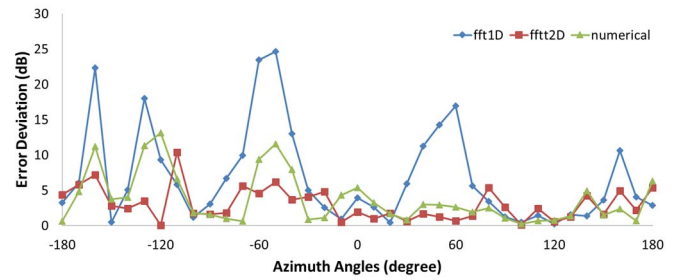


Fig. 5. Different plots between far-field and near-field measurement results

Those results show that the FFT-2D can solve the problem more correct than the FFT-1D. This condition can be verified with two equations that solved with the two calculation methods, i.e. Equations (7) and (8). Those equations are in two levels of integral form which are more correct to be solved with either FFT-2D and numerical method than the FFT-1D. The discrepancies between FFT-2D and the numerical calculation

shows better calculation accuracy conducted with the FFT-2D compared to the numerical method. This is because the equations have an appropriate form to be solved with the FFT method, therefore, can provide accurate and effective calculation process. Meanwhile, the deviations of the FFT-2D and the numerical relative to the measurement results are probably affected by the number of sampling in the near-field measurement and/or precision of the measurement setup both in the near-field and the far-field measurement.

V. CONCLUSION

Transformation results show that average deviations of one dimension Fast-Fourier Transform, two dimensions Fast-Fourier Transform and the numerical method are 6.83 dB, 3.04 dB, and 3.77 dB, respectively. These results show the minimum discrepancies between the transformed far-field and the expected far-field obtained by using the two dimensions Fast-Fourier Transform. This method has shown near-field measurement and transformation can predict the far-field radiation characteristics.

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